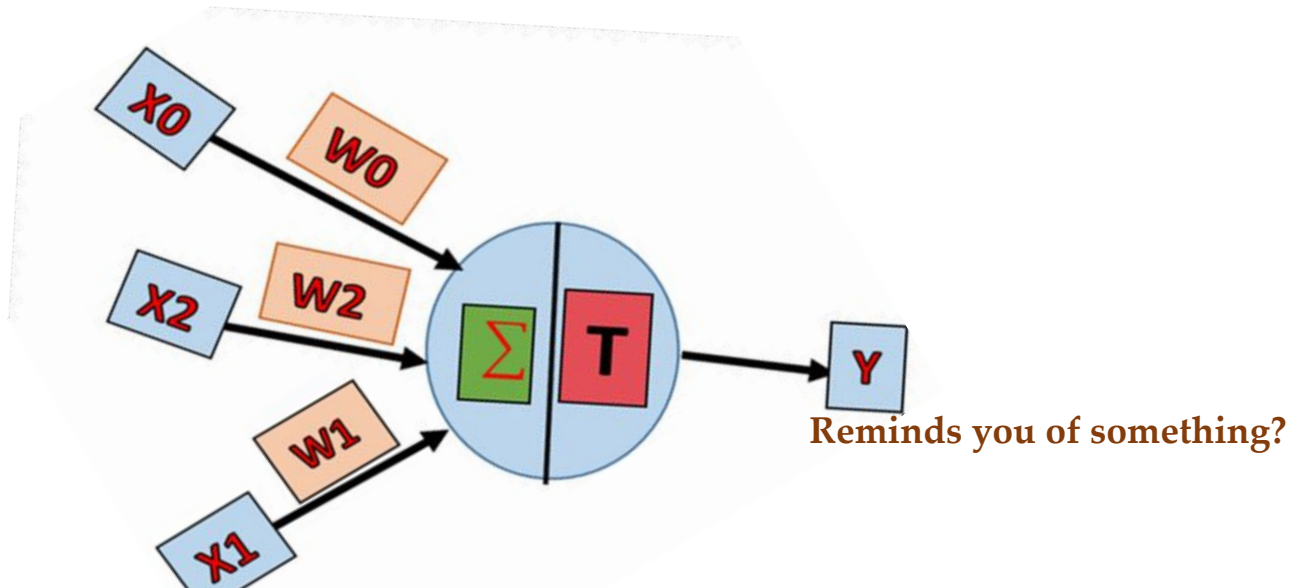


Artificial Intelligence and Machine Learning

6CS012 - Week 03 –Sunita Parajuli

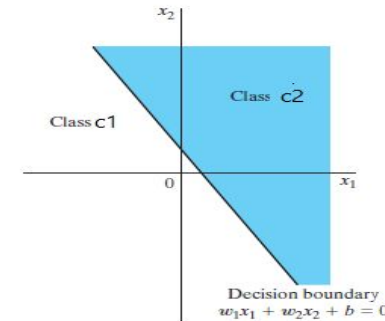
The Perceptron Model.

- The **Perceptron**, introduced by **Frank Rosenblatt** and later corrected by **Minsky and Papert**, extends the **McCulloch-Pitts neuron**
 - by incorporating **learnable numerical weights** and
 - an **adaptive learning process**.
 - It serves as a **Linear Binary Classifier**, dividing data into two categories based on a linear decision boundary.



Perceptron Algorithm

- The perceptron algorithm is built around a *single neuron* and it is limited to performing pattern classification with only two classes (for example: 0 and 1 class or 1 and -1 class, called as binary class)
- The perceptron is the simplest form of a neural network used for the classifying linearly separable patterns. Patterns that can be separated by a hyperplane are called linearly separable patterns.



- Perceptron can be trained to show behavior of AND function and OR function because both are linearly separable.

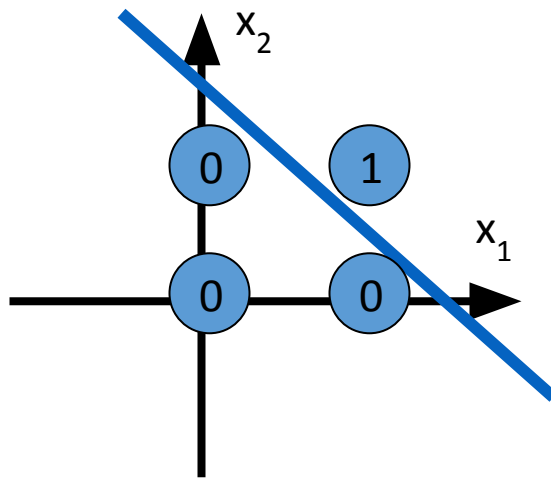


Fig: AND Function

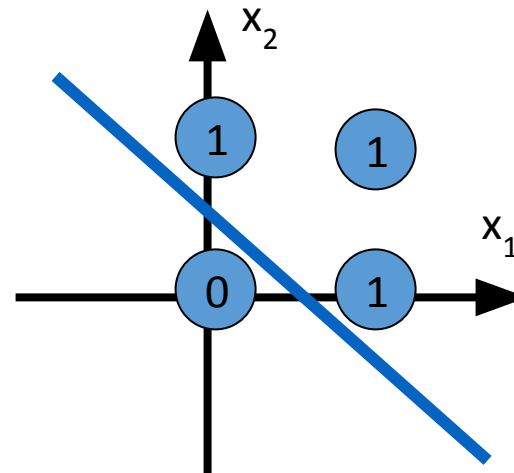


Fig: OR Function

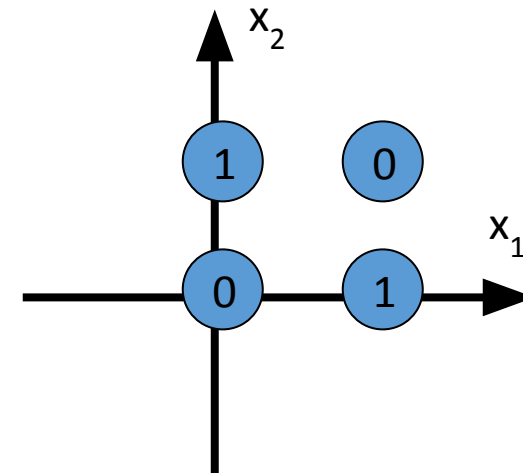


Fig: XOR Function

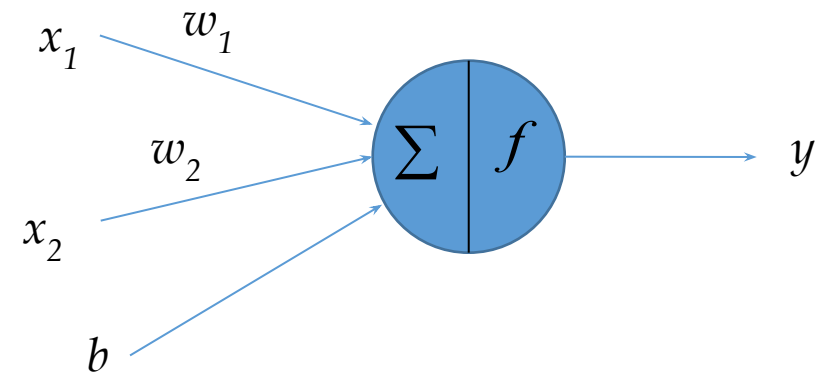
Perceptron Learning Algorithm

1. Initialize all weights (w) and bias (b) to zero
2. For each training vector x and actual target ' t ' (**desired output**)
3. Compute output y from y_{in} using activation function ' f ':

$$y_{in} = b + \sum_{i=1}^n w_i x_i \quad y = f(y_{in})$$

1. Update weights as: $w_i = w_i + \alpha(t - y)x_i$ for $i = 1$ to n
2. Update bias as: $b = b + \alpha(t - y)$
3. Repeat updating weight and bias until desired output.

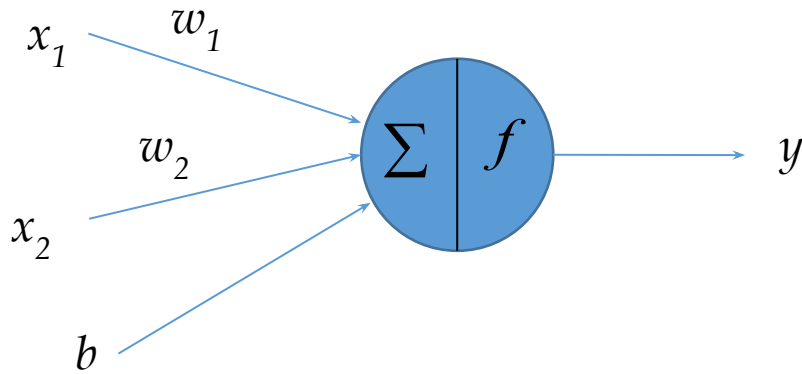
Note: α is learning rate, $0 \leq \alpha \leq 1$



Realizing AND gate using One Neuron (Perceptron Algorithm)

Example

Train the following perceptron by using given training set



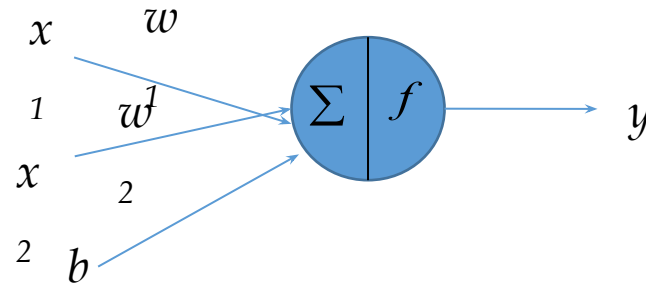
x_1	x_2	Target (t) $x_1 \text{ AND } x_2$
1	1	1
1	-1	-1
-1	1	-1
-1	-1	-1

and given activation function is

$$f(v) = \begin{cases} 1 & \text{if } v > 0 \\ -1 & \text{if } v < 0 \\ 0 & \text{if } v = 0 \end{cases} \text{ here, } v = w_1 * x_1 + w_2 * x_2 + b$$

Solution: $\alpha=1$

Iteration#1



Input(x_1, x_2, target)

$$v = w_1 * x_1 + w_2 * x_2 + b$$

$$w_2 = w_2 + \alpha * (\text{target} - y) * x_2$$

$$b = b + \alpha * (\text{target} - y)$$

Input	w_1 (old)	w_2 (old)	b (old)	v	y	w_1 (new)	w_2 (new)	b (new)
(1,1,1)	0	0	0	0	0	$0+1*1*1=1$	$0+1*1*1=1$	$0+1*1=1$
(1,-1,-1)	1	1	1	1	1	$1+1*-2*1=-1$	$1+1*-2*-1=3$	$1+1*-2=-1$
(-1,1,-1)	-1	3	-1	3	1	$-1+1*-2*-1=1$	$3+1*-2*1=1$	$-1+1*-2=-3$
(-1,-1,-1)	1	1	-3	-5	-1	$1+1*0*-1=1$	$1+1*0*-1=1$	$-3+1*0=-3$

$y = f(v)$ i.e. activation function applied to v

Let,

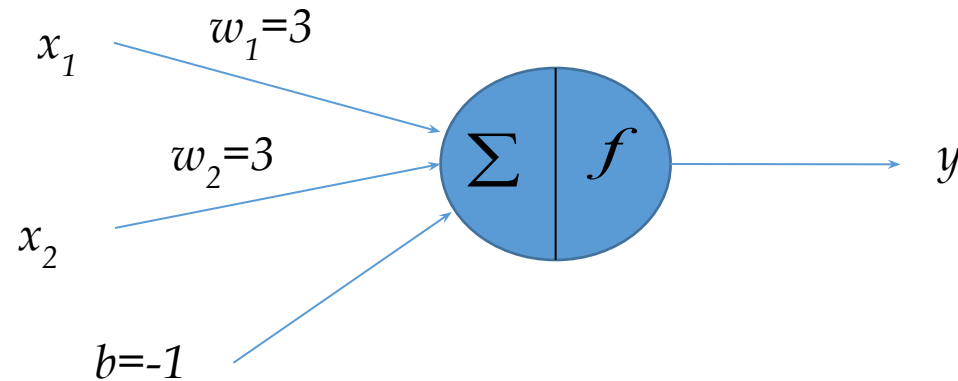
$$f(v) = \begin{cases} 1 & \text{if } v > 0 \\ -1 & \text{if } v < 0 \\ 0 & \text{if } v = 0 \end{cases}$$

$$w_1 = w_1 + \alpha * (\text{target} - y) * x_1$$

Iteration #2

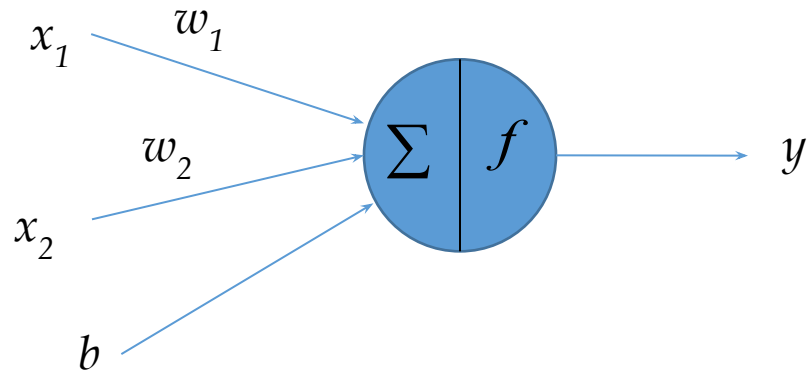
<i>Input</i>	w_1 (old)	w_2 (old)	b (old)	v	y	w_1 (new)	w_2 (new)	b (new)
(1,1,1)	1	1	-3	-1	-1	$1+1*2*1=3$	$1+1*2*1=3$	$-3+1*2=-1$
(1,-1,-1)	3	3	-1	-1	-1	$3+1*0*1=3$	$3+1*0*-1=3$	$-1+1*0=-1$
(-1,1,-1)	3	3	-1	-1	-1	$2+1*0*-1=3$	$3+1*0*1=3$	$-1+1*0=-1$
(-1,-1,-1)	3	3	-1	-7	-1	$3+1*0*-1=3$	$3+1*0*-1=3$	$-1+1*0=-1$

Thus, Final Neuron is:



No update in
weights and
bias so
solution is
obtained

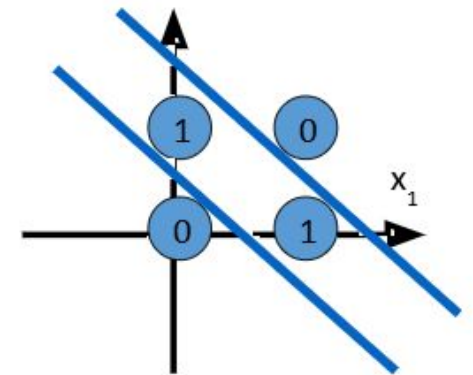
Try Realizing XOR gate using the perceptron



x_1	x_2	Target (t) <i>x1 AND x2</i>
1	1	-1
1	-1	1
-1	1	1
-1	-1	-1

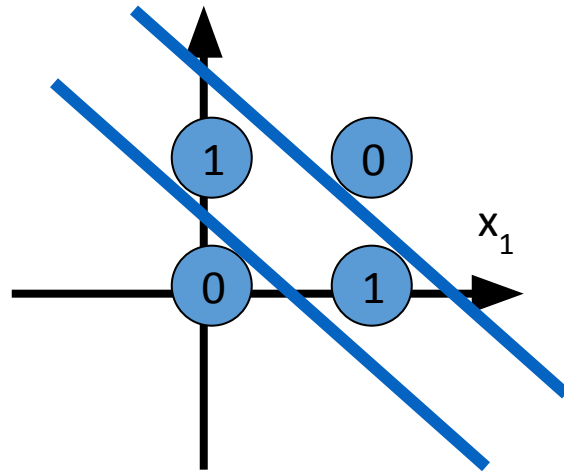
and given activation function is

$$f(v) = \begin{cases} 1 & \text{if } v > 0 \\ -1 & \text{if } v < 0 \\ 0 & \text{if } v = 0 \end{cases} \text{ here, } v = w_1 * x_1 + w_2 * x_2 + b$$

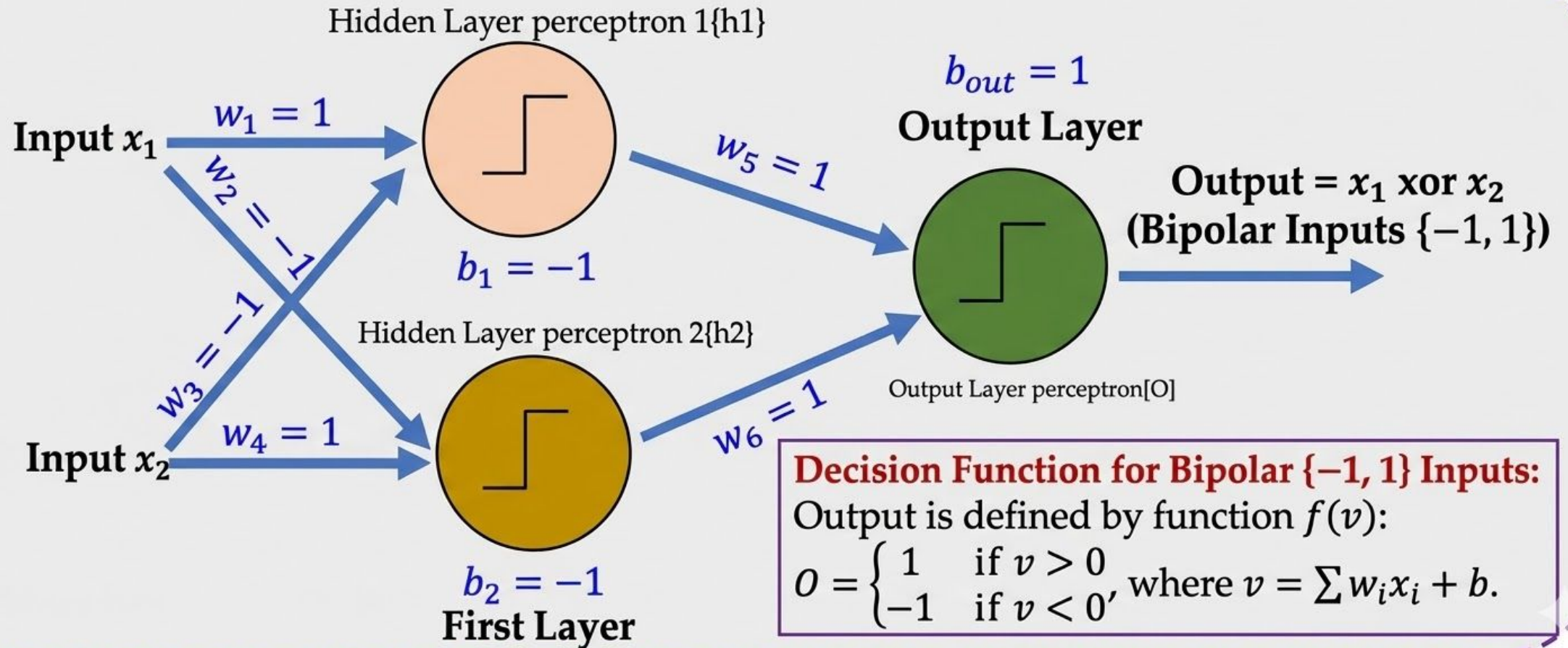


XOR Problem

- single layer perceptron can be trained to classify patterns that is **linearly separable**.
- However, XOR function is non-linearly separable as shown in the figure given below. Thus, we can not train single layer perceptron for XOR function. This, problem is called XOR Problem.



Solving XOR With Multilayer Perceptron



Solving XOR with multi layer perceptron

Inputs (x1,x2)	Hidden 1 $v1(1x1-1x2-1)$	h1 Outputf(v1)	Hidden 2 $v2(-1x1+1x2-1)$	h2 Outputf (v2)	Output $vout(1h1+1h2+1)$	Final Outputf (vout)	Target
(1, 1)	$1(1) - 1(1) - 1 = -1$	-1	$-1(1) + 1(1) - 1 = -1$	-1	$1(-1) + 1(-1) + 1 = -1$	-1	-1
(1, -1)	$1(1) - 1(-1) - 1 = 1$	1	$-1(1) + 1(-1) - 1 = -3$	-1	$1(1) + 1(-1) + 1 = 1$	1	1
(-1, 1)	$1(-1) - 1(1) - 1 = -3$	-1	$-1(-1) + 1(1) - 1 = 1$	1	$1(-1) + 1(1) + 1 = 1$	1	1
(-1, -1)	$1(-1) - 1(-1) - 1 = -1$	-1	$-1(-1) + 1(-1) - 1 = -1$	-1	$1(-1) + 1(-1) + 1 = -1$	-1	-1

Any Questions?